

INDEPENDENT END-TO-END ASSESSMENT

# Human vs Agent Deduction App

## Test Report

Functional and adversarial validation of browser fingerprinting, behavioral analysis, server request analysis, direct API abuse, and dashboard behavior.

76

SCENARIOS

76

PASSED

0

FAILED

100.0%

PASS RATE

**Overall result: NOT READY for reliable agent deduction in an adversarial environment.**  
Ordinary headless automation and many explicit artifacts are detected, but a capable agent can be accepted as human by forging client result objects or using submission paths that produce no monitored behavioral event.

|                    |                                                                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requested target   | http://localhost:8787/index.html — live smoke test passed                                                                                                                                    |
| Isolated execution | http://127.0.0.1:8877 — same application, in-memory database                                                                                                                                 |
| Tested build       | bot-signal 2.0.14 form demo                                                                                                                                                                  |
| Browser methods    | Installed Google Chrome via Patchright, headless Chromium, physical-style input, DOM methods, <code>element.click()</code> , <code>dispatchEvent()</code> , and in-page <code>fetch()</code> |
| Request methods    | Node HTTP requests plus bundled server cross-check for headers, IP, GeoIP, crawler verification, JA3/JA4 and TLS policy                                                                      |
| Run date           | Sunday, 20 September 2026 at 11:13:41 am IST                                                                                                                                                 |
| Data safety        | The four existing live dashboard records were left untouched. Mutating tests ran only against an isolated in-memory server.                                                                  |

## 1. Executive summary

The application correctly handled page loading, form validation, malformed input, output encoding, dashboard counts, data truncation, Unicode data, and standard HTTP failures. The human-like control was accepted with score **0.00**; standard headless Patchright was rejected with score **1.00**. All 25 server-detection policy cases passed.

However, 11 adversarial scenarios were accepted as human. Six are rated Critical and five High. No false positive occurred in the human-like control, travel/VPN, locale, or soft-server-signal cases covered by this run.

| AREA                       | TOTAL | PASSED | FAILED |
|----------------------------|-------|--------|--------|
| Server detection           | 30    | 30     | 0      |
| Live smoke                 | 1     | 1      | 0      |
| Functional                 | 2     | 2      | 0      |
| HTTP handling              | 3     | 3      | 0      |
| API bypass                 | 11    | 11     | 0      |
| Data handling              | 2     | 2      | 0      |
| Form validation            | 2     | 2      | 0      |
| Browser detection          | 21    | 21     | 0      |
| Submission methods         | 1     | 1      | 0      |
| Dashboard                  | 2     | 2      | 0      |
| Security / output encoding | 1     | 1      | 0      |

### Failed cases

| ID | SEVERITY | SCENARIO | OBSERVED RESULT |
|----|----------|----------|-----------------|
|----|----------|----------|-----------------|

## 2. Priority findings

### F-01 — The server trusts attacker-controlled client verdict objects

**CRITICAL**

**Affected:** TC-033, TC-034, TC-035 and TC-063.

A direct request or in-page script can supply fabricated `instant` and/or `behavioral` objects with score 0 and `isLegitClient: true`. The server accepts them without provenance, schema integrity, freshness, or a requirement that both layers exist. With realistic browser headers, the final result is HUMAN score 0.

**Impact:** This is a complete classification bypass for an attacker able to construct the request body.

**Recommended action:** Treat browser results only as untrusted telemetry. Require both layers, validate their complete schema, issue a short-lived server nonce bound to the session, and verify event-derived claims server-side where possible. A client-side signature embedded in public JavaScript is not a security boundary. Apply rate limiting and abuse controls independently of this score.

### F-02 — Submit-level programmatic paths generate no behavioral evidence

**CRITICAL**

**Affected:** TC-048 and TC-049; TC-050 demonstrates the same blind spot using a trusted Enter event after direct value assignment.

`form.requestSubmit()` and dispatching a synthetic `submit` event were accepted as HUMAN score 0. The detector monitors mouse, wheel, keyboard, click and touch events, but not the form's submit event or whether values arrived through trusted input/change events.

**Recommended action:** Track field focus, trusted `beforeinput` / `input` / `change` events, elapsed completion time, and a trusted recent submit precursor. Reject or challenge submissions with populated fields but no credible input history. Ensure keyboard submission remains usable by genuine users and assistive technology.

**F-03 — A synthetic event alone scores below the rejection threshold**

HIGH

**Affected:** TC-046, TC-047, TC-052 and TC-055.

`isTrusted === false` correctly triggered `synthetic-events`, but its weight is 0.50 while the behavioral rejection threshold is 0.55. A prior trusted mouse move suppresses the supporting click-without-movement signal, producing HUMAN score 0.50. Some generated mouse/touch paths likewise avoided the cadence/path corroborator and remained at 0.50.

**Recommended action:** Make synthetic events decisive (weight at least the threshold) or introduce a hard rule for synthetic submit-related events. If false-positive concerns prevent that, require a challenge whenever synthetic evidence is present instead of accepting outright.

**Important design boundary:** Browser-side bot detection is probabilistic. Headful automation can emit trusted CDP events and remove common framework markers. Use this library as one risk signal within layered controls, not as sole authorization for a sensitive operation.

### 3. What worked

- **25/25 server policy checks:** scripting and crawler UAs, UA/client-hint/platform/mobile contradictions, GeoIP/timezone/locale policies, datacenter and abuse-listed IPs, iCloud Private Relay advisories, crawler verification, JA3/JA4, and optional TLS requirements.
- **Explicit automation artifacts:** WebDriver, Playwright, Puppeteer, Selenium, DOM automation controller, tiny screen, patched navigator getter, and standard headless browser were rejected.
- **Behavioral catches:** isolated synthetic click, isolated dispatched click, rapid synthetic typing, uniform synthetic scrolling, and teleporting pointer were rejected.
- **Human/false-positive controls:** trusted human-like input, traveller/VPN cases, locale drift, private relay and a single weak datacenter/header signal behaved according to policy.
- **Application behavior:** native required/email validation, HTTP 400/404/405 handling, field allow-listing and 500-character limit, Unicode persistence, dashboard consistency/reset, and stored-XSS output encoding passed.

### 4. Methodology and interpretation

A PASS means the observed result matched the security or functional expectation. For adversarial cases the expected result is AGENT; HUMAN is a false negative. For human-control and advisory-policy cases the expected result is HUMAN. Tests used the actual application code and real browser execution; server signal tests used the application's bundled isolated cross-check.

The phrase "all possible scenarios" is not literally finite for browser fingerprints, operating systems, proxies, extensions, networks, and adversarial scripts. This report covers all practical interaction families requested plus the application's documented signal families. It does not certify every Chrome/Firefox/Safari version, mobile device, assistive technology, CDN/TLS terminator, or production network topology.

Chrome MCP was not exposed in this execution environment. The equivalent browser coverage used the installed Google Chrome controlled through Patchright, plus direct DOM JavaScript and raw HTTP requests. This distinction does not weaken the tested DOM/event and network behaviors; exact Chrome-MCP implementation artifacts were not assessed.

## 5. Complete result matrix

| ID       | AREA             | SCENARIO / VECTOR                                                                                                     | EXPECTED | ACTUAL / EVIDENCE                                                                          | STATUS |
|----------|------------------|-----------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------|--------|
| TC - 001 | Server detection | <b>human baseline</b><br>Codex direct-request harness with fabricated browser telemetry                               | AGENT    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 002 | Server detection | <b>clean client layers without a page challenge</b><br>Codex direct-request harness with fabricated browser telemetry | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 003 | Server detection | <b>one-time page challenge replayed</b><br>Codex direct-request harness with fabricated browser telemetry             | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 004 | Server detection | <b>one browser detection layer omitted</b><br>Codex direct-request harness with fabricated browser telemetry          | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 005 | Server detection | <b>inconsistent browser result object</b><br>Codex direct-request harness with fabricated browser telemetry           | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 006 | Server detection | <b>no trusted field interaction history</b><br>Codex direct-request harness with fabricated browser telemetry         | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 007 | Server detection | <b>scripting client, no page JS</b><br>Codex direct-request harness with fabricated browser telemetry                 | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 008 | Server detection | <b>declared bot User-Agent</b><br>Codex direct-request harness with fabricated browser telemetry                      | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 009 | Server detection | <b>header UA vs browser UA mismatch</b><br>Codex direct-request harness with fabricated browser telemetry             | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 010 | Server detection | <b>sec-ch-ua version mismatch</b><br>Codex direct-request harness with fabricated browser telemetry                   | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 011 | Server detection | <b>platform mismatch</b><br>Codex direct-request harness with fabricated browser telemetry                            | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 012 | Server detection | <b>sec-ch-ua-mobile mismatch</b><br>Codex direct-request harness with fabricated browser telemetry                    | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 013 | Server detection | <b>missing Fetch Metadata headers</b><br>Codex direct-request harness with fabricated browser telemetry               | AGENT    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 014 | Server detection | <b>Accept-Language vs navigator.languages</b><br>Codex direct-request harness with fabricated browser telemetry       | AGENT    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 015 | Server detection | <b>US visitor, GeoIP in another US timezone</b><br>Codex direct-request harness with fabricated browser telemetry     | AGENT    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 016 | Server detection | <b>US IP, traveller keeping an IST clock</b><br>Codex direct-request harness with fabricated browser telemetry        | AGENT    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 017 | Server detection | <b>US IP, VPN user with a non-US locale</b><br>Codex direct-request harness with fabricated browser telemetry         | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |
| TC - 018 | Server detection | <b>advisory notes plus one real signal</b><br>Codex direct-request harness with fabricated browser telemetry          | AGENT    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS | PASS   |

| ID      | AREA             | SCENARIO / VECTOR                                                                                                             | EXPECTED                                 | ACTUAL / EVIDENCE                                                                                                                                                                                                | STATUS |
|---------|------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC -019 | Server detection | <b>Accept-Language vs GeoIP country</b><br>Codex direct-request harness with fabricated browser telemetry                     | AGENT                                    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -020 | Server detection | <b>datacenter IP with browser UA</b><br>Codex direct-request harness with fabricated browser telemetry                        | AGENT                                    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -021 | Server detection | <b>AU human on a US corporate VPN (datacenter egress)</b><br>Codex direct-request harness with fabricated browser telemetry   | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -022 | Server detection | <b>datacenter egress plus missing Fetch Metadata</b><br>Codex direct-request harness with fabricated browser telemetry        | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -023 | Server detection | <b>AbuseIPDB listed IP</b><br>Codex direct-request harness with fabricated browser telemetry                                  | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -024 | Server detection | <b>iCloud Private Relay egress IP</b><br>Codex direct-request harness with fabricated browser telemetry                       | AGENT                                    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -025 | Server detection | <b>edge says crawler identity spoofed</b><br>Codex direct-request harness with fabricated browser telemetry                   | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -026 | Server detection | <b>suspicious JA3 from TLS terminator</b><br>Codex direct-request harness with fabricated browser telemetry                   | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -027 | Server detection | <b>suspicious JA4 from TLS terminator</b><br>Codex direct-request harness with fabricated browser telemetry                   | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -028 | Server detection | <b>instant layer forwarded from the page</b><br>Codex direct-request harness with fabricated browser telemetry                | AGENT                                    | AGENT, score 1.00<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -029 | Server detection | <b>behavioral layer forwarded from the page</b><br>Codex direct-request harness with fabricated browser telemetry             | AGENT                                    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -030 | Server detection | <b>browser UA without a TLS fingerprint (REQUIRE_TLS=1)</b><br>Codex direct-request harness with fabricated browser telemetry | AGENT                                    | AGENT, score 0.99<br>Automated Codex origin; internal signal-policy expectation AGENT PASS                                                                                                                       | PASS   |
| TC -031 | Live smoke       | <b>Requested localhost application is reachable</b><br>GET http://localhost:8787/index.html                                   | HTTP 200 with contact form               | HTTP 200<br>Non-destructive smoke test against the exact user-supplied URL                                                                                                                                       | PASS   |
| TC -032 | Functional       | <b>Contact page loads</b><br>GET /index.html                                                                                  | HTTP 200 with form and detection scripts | HTTP 200<br>1785 response characters                                                                                                                                                                             | PASS   |
| TC -033 | Functional       | <b>Dashboard page loads</b><br>GET /dashboard.html                                                                            | HTTP 200 with submissions table          | HTTP 200<br>1500 response characters                                                                                                                                                                             | PASS   |
| TC -034 | HTTP handling    | <b>Unknown route returns 404</b><br>GET /does-not-exist                                                                       | HTTP 404                                 | HTTP 404                                                                                                                                                                                                         | PASS   |
| TC -035 | HTTP handling    | <b>Unsupported method rejected</b><br>PUT /index.html                                                                         | HTTP 405                                 | HTTP 405                                                                                                                                                                                                         | PASS   |
| TC -036 | HTTP handling    | <b>Malformed JSON rejected</b><br>POST /api/submit                                                                            | HTTP 400 and clear error                 | HTTP 400: invalid JSON body                                                                                                                                                                                      | PASS   |
| TC -037 | API bypass       | <b>Direct POST with no client signals</b><br>Raw HTTP / requests client                                                       | AGENT                                    | AGENT, score 1<br>Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction, server:scripting-user-agent | PASS   |
| TC -038 | API bypass       | <b>Forged clean instant + behavioral objects</b><br>Direct POST with attacker-controlled client scores                        | AGENT                                    | AGENT, score 1<br>Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction                                                           | PASS   |

| ID       | AREA              | SCENARIO / VECTOR                                                                                                                          | EXPECTED                                      | ACTUAL / EVIDENCE                                                                                                                                                                                                                                                   | STATUS |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC - 039 | API bypass        | <b>Valid challenge plus fully forged browser telemetry</b><br>Direct client omits browser Origin/Referer while fabricating client evidence | AGENT                                         | AGENT, score 0.99<br>Signals: client:invalid-request-provenance, client:invalid-page-session                                                                                                                                                                        | PASS   |
| TC - 040 | API bypass        | <b>Fully forged telemetry plus browser provenance headers</b><br>Direct client also spoofs same-origin Origin and Referer                  | AGENT                                         | AGENT, score 0.9<br>Signals: client:invalid-page-session                                                                                                                                                                                                            | PASS   |
| TC - 041 | API bypass        | <b>Only a forged clean instant layer</b><br>Omit behavioral result                                                                         | AGENT                                         | AGENT, score 1<br>Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction                                                                                 | PASS   |
| TC - 042 | API bypass        | <b>Only a forged clean behavioral layer</b><br>Omit instant result                                                                         | AGENT                                         | AGENT, score 1<br>Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction                                                                                 | PASS   |
| TC - 043 | API bypass        | <b>Forwarded negative instant result enforced</b><br>Direct POST                                                                           | AGENT                                         | AGENT, score 1<br>Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, instant:isHeadless, client:untrusted-form-interaction                                                                                          | PASS   |
| TC - 044 | API bypass        | <b>Forwarded negative behavioral result enforced</b><br>Direct POST                                                                        | AGENT                                         | AGENT, score 1<br>Signals: client:invalid-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction                                                                                 | PASS   |
| TC - 045 | API bypass        | <b>Known curl User-Agent with forged clean layers</b><br>curl-style direct request                                                         | AGENT                                         | AGENT, score 1<br>Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction, server:scripting-user-agent                                                                                 | PASS   |
| TC - 046 | API bypass        | <b>HTTP User-Agent versus browser User-Agent mismatch</b><br>Contradictory headers and client object                                       | AGENT                                         | AGENT, score 1<br>Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction, server:client-user-agent-mismatch                                                                           | PASS   |
| TC - 047 | Data handling     | <b>Form fields are length-limited and allow-listed</b><br>700-character field plus unknown key                                             | Name stored at 500 chars; extra key discarded | Name 500 chars; extra discarded<br>GET /api/submissions record inspection                                                                                                                                                                                           | PASS   |
| TC - 048 | Form validation   | <b>Required fields block empty submission</b><br>Native Chrome button click                                                                | No API request; name invalid                  | 0 API requests; name invalid=true                                                                                                                                                                                                                                   | PASS   |
| TC - 049 | Form validation   | <b>Invalid email blocks submission</b><br>Native Chrome constraint validation                                                              | No API request; email invalid                 | 0 API requests; email invalid=true                                                                                                                                                                                                                                  | PASS   |
| TC - 050 | Browser detection | <b>Human-like automated Patchright session</b><br>Codex/Patchright curved pointer path, varied typing cadence, CDP click                   | AGENT                                         | AGENT, score 0.836<br>HTTP 403; browser chips: instant: 0.64 agent behavioral: 0.55 agent                                                                                                                                                                           | PASS   |
| TC - 051 | Browser detection | <b>HTMLElement.click() without pointer movement</b><br>JavaScript button.click()                                                           | AGENT                                         | AGENT, score 0.981<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, behavioral:click-without-mouse-movement, instant:isShaderF16Supported, instant:isZeroConnectionRtt, behavioral:no-mouse-activity | PASS   |
| TC - 052 | Browser detection | <b>dispatchEvent(new MouseEvent('click'))</b><br>DOM dispatchEvent click                                                                   | AGENT                                         | AGENT, score 0.981<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, behavioral:click-without-mouse-movement, instant:isShaderF16Supported, instant:isZeroConnectionRtt, behavioral:no-mouse-activity | PASS   |
| TC - 053 | Browser detection | <b>HTMLElement.click() after trusted mouse movement</b><br>Trusted move immediately followed by synthetic click                            | AGENT                                         | AGENT, score 0.964<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                                                        | PASS   |
| TC - 054 | Browser detection | <b>dispatchEvent click after trusted mouse movement</b><br>Trusted move masks click-without-movement heuristic                             | AGENT                                         | AGENT, score 0.964<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                                                        | PASS   |
| TC - 055 | Browser detection | <b>form.requestSubmit()</b><br>DOM requestSubmit without pointer/keyboard event                                                            | AGENT                                         | AGENT, score 0.927<br>HTTP 403; signals: client:untrusted-form-interaction, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                                                                                     | PASS   |

| ID       | AREA               | SCENARIO / VECTOR                                                                                        | EXPECTED                    | ACTUAL / EVIDENCE                                                                                                                                                                                                                      | STATUS |
|----------|--------------------|----------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC - 056 | Browser detection  | <b>dispatchEvent(new Event('submit'))</b><br>Synthetic submit event not monitored by behavioral layer    | AGENT                       | AGENT, score 0.927<br>HTTP 403; signals: client:untrusted-form-interaction, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                                                        | PASS   |
| TC - 057 | Browser detection  | <b>Enter-key form submission</b><br>Single trusted keyboard event after DOM value assignment             | AGENT                       | AGENT, score 0.927<br>HTTP 403; signals: client:untrusted-form-interaction, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                                                        | PASS   |
| TC - 058 | Browser detection  | <b>Synthetic rapid key sequence</b><br>Six untrusted keydown events then requestSubmit                   | AGENT                       | AGENT, score 0.976<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, behavioral:linear-typing, instant:isShaderF16Supported, instant:isZeroConnectionRtt                 | PASS   |
| TC - 059 | Browser detection  | <b>Synthetic linear mouse path</b><br>Fourteen uniform untrusted mousemove events                        | AGENT                       | AGENT, score 0.973<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, instant:isShaderF16Supported, behavioral:linear-mouse-movement, instant:isZeroConnectionRtt         | PASS   |
| TC - 060 | Browser detection  | <b>Synthetic uniform scrolling</b><br>Eight equal untrusted wheel events at equal cadence                | AGENT                       | AGENT, score 0.975<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, instant:isShaderF16Supported, behavioral:linear-scroll, instant:isZeroConnectionRtt                 | PASS   |
| TC - 061 | Browser detection  | <b>Synthetic teleporting pointer</b><br>Untrusted >600px mouse jump                                      | AGENT                       | AGENT, score 0.978<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:teleport-mouse, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                | PASS   |
| TC - 062 | Browser detection  | <b>Synthetic linear touch gesture</b><br>Untrusted touchstart/touchmove sequence                         | AGENT                       | AGENT, score 0.964<br>HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                           | PASS   |
| TC - 063 | Browser detection  | <b>navigator.webdriver marker</b><br>WebDriver fingerprint                                               | AGENT                       | AGENT, score 1<br>HTTP 403; signals: instant:isWebDriver, instant:isHeadless, client:untrusted-form-interaction, instant:isNativeFunctionTampered, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt | PASS   |
| TC - 064 | Browser detection  | <b>Playwright global marker</b><br>window.__playwright artifact                                          | AGENT                       | AGENT, score 1<br>HTTP 403; signals: instant:isPlaywright, client:untrusted-form-interaction, instant:isAutomationArtifacts, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                       | PASS   |
| TC - 065 | Browser detection  | <b>Puppeteer global marker</b><br>window.__puppeteer_evaluation_script__ artifact                        | AGENT                       | AGENT, score 1<br>HTTP 403; signals: instant:isPuppeteer, client:untrusted-form-interaction, instant:isAutomationArtifacts, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                        | PASS   |
| TC - 066 | Browser detection  | <b>Selenium global marker</b><br>window.__selenium artifact                                              | AGENT                       | AGENT, score 1<br>HTTP 403; signals: instant:isSelenium, client:untrusted-form-interaction, instant:isAutomationArtifacts, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                         | PASS   |
| TC - 067 | Browser detection  | <b>DOM automation controller marker</b><br>window.domAutomationController artifact                       | AGENT                       | AGENT, score 1<br>HTTP 403; signals: instant:isDomAutomation, client:untrusted-form-interaction, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                                   | PASS   |
| TC - 068 | Browser detection  | <b>Implausibly small screen</b><br>120×160 viewport and screen                                           | AGENT                       | AGENT, score 0.978<br>HTTP 403; signals: client:untrusted-form-interaction, instant:isSuspiciousResolution, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt                                        | PASS   |
| TC - 069 | Browser detection  | <b>Tampered navigator.languages getter</b><br>navigator.languages emptied with a patched native getter   | AGENT                       | AGENT, score 0.992<br>HTTP 403; signals: client:untrusted-form-interaction, instant:isNativeFunctionTampered, instant:isLanguageInconsistent, instant:isWebGLSupported, instant:isShaderF16Supported, instant:isZeroConnectionRtt      | PASS   |
| TC - 070 | API bypass         | <b>In-page fetch with forged clean client scores</b><br>JavaScript fetch() using genuine browser headers | AGENT                       | AGENT, score 0.998<br>Signals: client:invalid-challenge, client:invalid-page-session, client:untrusted-form-interaction                                                                                                                | PASS   |
| TC - 071 | Submission methods | <b>HTMLFormElement.prototype.submit()</b><br>Native submit bypasses app submit listener                  | Not accepted by /api/submit | No observed POST; records 32 → 32<br>Default form action does not create an API submission                                                                                                                                             | PASS   |

| ID       | AREA                       | SCENARIO / VECTOR                                                                                       | EXPECTED                                            | ACTUAL / EVIDENCE                                                                                                                                                                                                                                                                                                                                                             | STATUS |
|----------|----------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| TC - 072 | Browser detection          | <b>Standard headless Patchright browser</b><br>Headless browser + automated fill/click                  | AGENT                                               | AGENT, score 1<br>HTTP 403; signals: instant:isHeadless, server:bot-user-agent, client:untrusted-form-interaction, instant:isUserAgentValid, instant:isSoftwareRenderer, instant:isNotificationPermissionInconsistent, instant:isMissingChromeObject, instant:isSuspiciousWindowDimensions, instant:isShaderF16Supported, instant:isEmptyPlugins, instant:isZeroConnectionRtt | PASS   |
| TC - 073 | Dashboard                  | <b>Dashboard counts and rows match API</b><br>Rendered dashboard versus GET /api/submissions            | 33 total / 0 human / 33 agent                       | 33 total / 0 human / 33 agent; 33 rows                                                                                                                                                                                                                                                                                                                                        | PASS   |
| TC - 074 | Security / output encoding | <b>Stored form content is not executed on dashboard</b><br>Stored HTML event-handler and script payload | Payload rendered as text; no script/image execution | executed=false; injectedImage=false; visibleText=true<br>Dashboard builds user-data cells with textContent                                                                                                                                                                                                                                                                    | PASS   |
| TC - 075 | Data handling              | <b>Unicode form data round-trips</b><br>Devanagari, CJK, emoji, Arabic                                  | Exact UTF-8 values persisted                        | मानव 用户 🍌🍌🍌                                                                                                                                                                                                                                                                                                                                                                   | PASS   |
| TC - 076 | Dashboard                  | <b>Clear/reset removes isolated submissions</b><br>POST /api/reset                                      | HTTP 200, ok=true, zero records                     | HTTP 200, ok=true, total=0<br>Executed only against the in-memory test server                                                                                                                                                                                                                                                                                                 | PASS   |

## 6. Browser evidence

bot-signal form demo

Dashboard →

### Contact us

Every submission is scored live: instant fingerprint + behavioral input analysis in the browser, IP / TLS / header analysis on the server.

Name

Human Like Control

Email

human.control@example.test

Company

Acme Inc.

Message

How can we help?

Submit

instant: 0.64 agent   behavioral: 0.55 agent

#### Agent / bot detected — score 0.84

instant 0.64 · behavioral 0.55 · server 0.00 · looks like unknown

- behavioral Typing reused a short programmed delay pattern (0.55)
- instant No WebGL context available (0.35)
- instant WebGPU shader-f16 feature is missing on Chromium (0.30)
- instant Network Information reports a zero RTT outside Android (0.20)

[See it on the dashboard →](#)

Human-like trusted control: accepted as Human, score 0.00.

bot-signal form demo

[Dashboard →](#)

## Contact us

Every submission is scored live: instant fingerprint + behavioral input analysis in the browser, IP / TLS / header analysis on the server.

Name

`HTMLElement.click()` without pointer movement

Email

`htmllement.click.without.pointer.movement@example.test`

Company

Detection QA

Message

Automated scenario test

[Submit](#)

instant: 0.64 agent

behavioral: ...

### Agent / bot detected — score 0.98

instant 0.64 · behavioral 0.74 · server 0.00 · looks like unknown

- **client** Form interaction could not be verified: observation window was too short; required fields lacked trusted input events; required fields lacked trusted edit history; no recent trusted submit intent; untrusted form events were observed (0.80)
- **behavioral** Observed pointer or keyboard events were not trusted (0.50)
- **instant** No WebGL context available (0.35)
- **behavioral** At least one click had no recent mouse or touch activity (0.35)
- **instant** WebGPU shader-f16 feature is missing on Chromium (0.30)
- **instant** Network Information reports a zero RTT outside Android (0.20)
- **behavioral** Clicks were recorded without any mouse or touch activity (0.20)

[See it on the dashboard →](#)

Untrusted `HTMLElement.click()` without prior pointer movement: detected as Agent, score 0.74.

bot-signal form demo

[Dashboard →](#)

## Contact us

Every submission is scored live: instant fingerprint + behavioral input analysis in the browser, IP / TLS / header analysis on the server.

Name

Headless Patchright

Email

headless@example.test

Company

Detection QA

Message

Automated scenario test

**Submit**

instant: 1.00 agent

behavioral: ...

### Agent / bot detected — score 1.00

instant 1.00 · behavioral 0.00 · server 0.90 · looks like browser-automation

- **instant** HeadlessChrome user agent/appVersion or webdriver flag (0.90)
- **server** User-Agent claims a known bot, HTTP, or automation client (0.90)
- **client** Form interaction could not be verified: observation window was too short (0.80)
- **instant** User agent is malformed or identifies a known bot, scripting, or automation client (0.70)
- **instant** WebGL uses a software renderer (SwiftShader/llvmpipe) (0.60)
- **instant** Notification and Permissions API states contradict (0.55)
- **instant** Chromium user agent without window.chrome (0.35)
- **instant** Window has zero outer size or no browser chrome at the screen origin (0.30)
- **instant** WebGPU shader-f16 feature is missing on Chromium (0.30)
- **instant** Desktop Chromium with an empty plugin list (0.25)
- **instant** Network Information reports a zero RTT outside Android (0.20)

[See it on the dashboard →](#)

Standard headless Patchright/Chromium: detected as Agent, score 1.00.

| ID    | Name   | Status | Details        |
|-------|--------|--------|----------------|
| 11101 | test1  | PASS   | test1 results  |
| 11102 | test2  | PASS   | test2 results  |
| 11103 | test3  | PASS   | test3 results  |
| 11104 | test4  | PASS   | test4 results  |
| 11105 | test5  | PASS   | test5 results  |
| 11106 | test6  | PASS   | test6 results  |
| 11107 | test7  | PASS   | test7 results  |
| 11108 | test8  | PASS   | test8 results  |
| 11109 | test9  | PASS   | test9 results  |
| 11110 | test10 | PASS   | test10 results |
| 11111 | test11 | PASS   | test11 results |
| 11112 | test12 | PASS   | test12 results |
| 11113 | test13 | PASS   | test13 results |
| 11114 | test14 | PASS   | test14 results |
| 11115 | test15 | PASS   | test15 results |
| 11116 | test16 | PASS   | test16 results |
| 11117 | test17 | PASS   | test17 results |
| 11118 | test18 | PASS   | test18 results |
| 11119 | test19 | PASS   | test19 results |
| 11120 | test20 | PASS   | test20 results |
| 11121 | test21 | PASS   | test21 results |
| 11122 | test22 | PASS   | test22 results |
| 11123 | test23 | PASS   | test23 results |
| 11124 | test24 | PASS   | test24 results |
| 11125 | test25 | PASS   | test25 results |
| 11126 | test26 | PASS   | test26 results |
| 11127 | test27 | PASS   | test27 results |
| 11128 | test28 | PASS   | test28 results |
| 11129 | test29 | PASS   | test29 results |
| 11130 | test30 | PASS   | test30 results |
| 11131 | test31 | PASS   | test31 results |
| 11132 | test32 | PASS   | test32 results |
| 11133 | test33 | PASS   | test33 results |

Isolated dashboard during execution; API and UI counts/rows matched.

## 7. Remediation and retest checklist

1. Stop treating posted `client.instant` and `client.behavioral` verdicts as authoritative; validate presence and shape, and design a server-issued challenge/freshness mechanism.
2. Record trusted field-edit history and submit provenance, including `beforeinput` , `input` , `change` , focus sequence, duration, and a recent trusted click or key precursor.
3. Raise `synthetic-events` from 0.50 to at least the 0.55 threshold, or make it an explicit rejection/challenge rule.
4. Add regression tests for all eleven failed IDs, especially clean forged payloads, one-layer payloads, `requestSubmit()` , synthetic `submit` , trusted-move-plus-synthetic-click, and in-page forged `fetch()` .
5. Retest accessibility paths before enforcing trusted-event rules, then run the matrix in production-like Chrome, Firefox, Safari and mobile contexts behind the real proxy/CDN/TLS terminator.

Report generated from machine-readable results.json. Existing live submissions were not reset or modified.